



The 'Superlaser' concept

Not as powerful as the Death Star, but we are now closer to merging multiple laser beams into a single powerful blast. <http://dgit.in/Superlaser>



The Last Jedi

The latest Star Wars VIII movie trailer is here. Luke Skywalker has finally spoken. <http://dgit.in/SWVIII>

Keyboards

HOW WE TESTED

Testing rig and setup: Earlier, processors weren't powerful enough to keep up with the load of multiple input devices pushing instructions down its throat while it was already busy processing other instructions from the OS. Considering today's hardware, your processor won't bottleneck even if you attach multiple keyboards or mice to it. However, to distance ourselves from any form of bottlenecks, we tested all the keyboards on the following rig:

- **Motherboard:** AsRock Z77 Extreme4
- **CPU:** Intel Core i7-3770K
- **GPU:** MSI Gaming X Radeon RX470
- **RAM:** Kingston HyperX DDR3 16GB
- **Monitor:** Dell Professional P2312H
- **Mouse:** Zowie FK2

Since we didn't want any inconsistencies between different test runs we didn't fiddle with the power or BIOS settings. Everything was set to default. The latest firmware was updated on every keyboard before testing. This involved either downloading it directly from the official website or through the software (if there was software support), and then upgrading. The keyboard was placed on a white hard plastic desk to test the anti-slip capability of the rubber or plastic feet on the bottom. Keyboards were connected to a USB 2.0 port to avoid any compatibility issues during the basic testing. Apart from that, the keyboards were also operated using USB 3.0 to verify whether there were any issues.

Features evaluation: The keyboards covered in this comparison test have a wide gamut of features. We made an effort throughout the test process to ensure all the features that are currently offered in keyboards were included in the scoring. Subsequently, the most important or necessary features for a gaming keyboard was given more weightage than those features which can be termed as luxury. For example, maximum n-key rollover was given more weightage than the number of lighting modes possible in the keyboard. Hardware features including the presence of USB and audio passthrough, Windows-key lock, multimedia buttons, etc., were covered to name a few. Keyboards that had software support were also given extra scores distributed over a set of features such as changing the polling rate, lighting modes, recording macros, etc. Accessories such as a keycap puller, extra keycaps, palm rest, carry pouches, etc., have also been scored.

Build quality: As all the keyboards are mechanical, most of them are sturdy. We made sure that every part and component of the keyboard was evaluated in terms of build quality. Here we didn't test how good a particular component was and instead, we scored based on the type of material it used. For example, keycaps are usually made from ABS or PBT plastic. To determine the material, we dipped a cotton swab in diluted Acetone and smudged it at the lower

edges of the keycaps. If the keycap is made of ABS, the coating or upper paint dissolves and gets smudged. For evaluation, we scored the usage of ABS and not how good the ABS material was on the keycap. Cherry MX switches have been the gold standard of mechanical switches and we also have a few knock-off switches competing in this test. We rated them based on their specifications since we don't have the necessary equipment to determine their durability over extended periods of wear.

Verdict: In our comparison tests, we give out three awards based on the final scores obtained after evaluating the features and build quality. Since we can't objectively determine the performance of the keyboard, the Best Performer award is a weighted sum of the features and build quality scores. This award will be given to the keyboard that obtains the highest score, revealing which keyboard is best in an overall aspect. The second award of Best Buy is given to the product with the highest score/rupee ratio. This award will help us determine which keyboard balances itself by offering good features and build quality at a proper price point. The Editor's Pick award is completely subjective and dependent on the experience of the reviewer. There can be several possibilities here and it might also be possible that one product wins Editor's Pick even though they might have won one of the two previously mentioned awards.

for keyboards and since their patent ended, several knock-offs have flooded the market. These knock-offs are comparatively cheaper and have drastically dropped the prices of mechanical keyboards. The basic "feel" of these switches are standard (denoted by colour) with every manufacturer (except Razer), certainly a good thing to avoid confusion. Although, with enough experience you will be able to observe the little nuances between them. The most common switches are Red – linear, Blue – tac-

tile and clicky, and Brown – tactile. Red switches are directed towards gaming, Blues are for typing whereas Brown stands at the sweet spot for both gamers and typists. Apart from preferences, Cherry switches are highly durable offering up to 50 million keystrokes that should last for a couple of decades. Yes, decades! One thing to note is that the durability is only for the key switches and not for the entire keyboard. Your keyboard's chassis or keycaps will start wearing off early but the switches

underneath will survive longer. Clearly, durability and a wide range of options in switch types in mechanical keyboards become a strong parameter for a gaming keyboard. Analog and optical technology for keyboard switches are also being considered. Both of them are based on the same mechanical technology, but the way input is registered by the keyboard is different. Going into detail will require its own in-depth story, but it's interesting to see new technologies coming up.

Coming over to the features that distinguish a gaming keyboard from a generic one, the level of n-key rollover is the most crucial. N-key rollover is used to denote how many keys can be registered simultaneously by the keyboard. The 'N' is replaced by a number, usually 3, 6, 10 or just N. '3' denotes that a maximum of three keys will be registered simultaneously which goes for any number replacing N. If N-key rollover or NKRO (sometimes full N-key rollover) is mentioned by the